

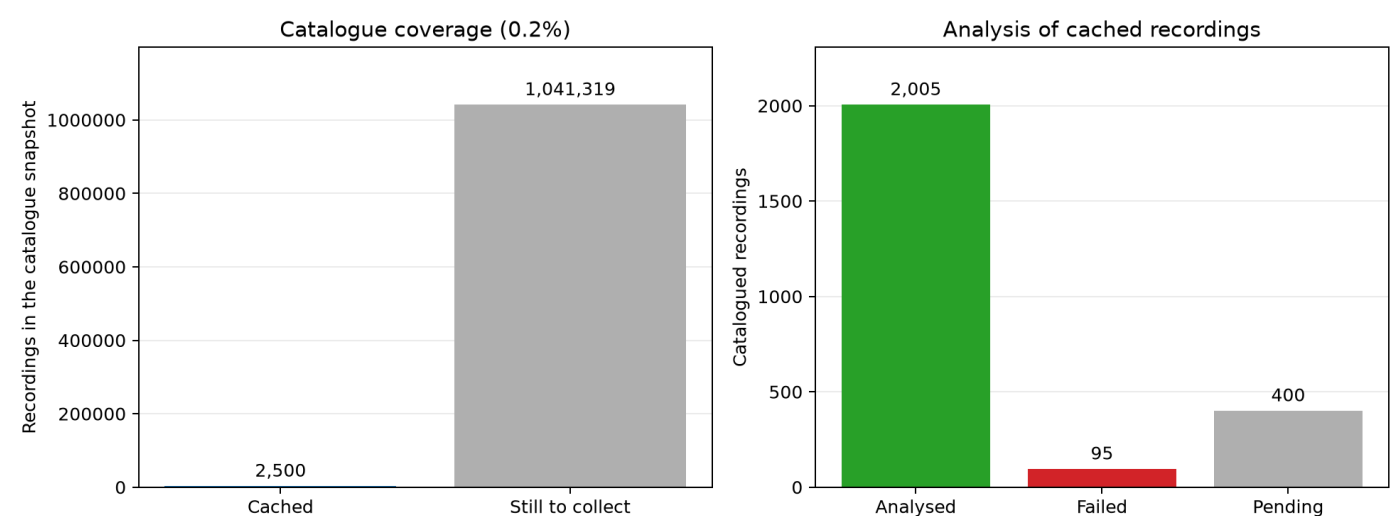
Xeno-canto robustness results

This notebook displays cumulative measurements from the database.

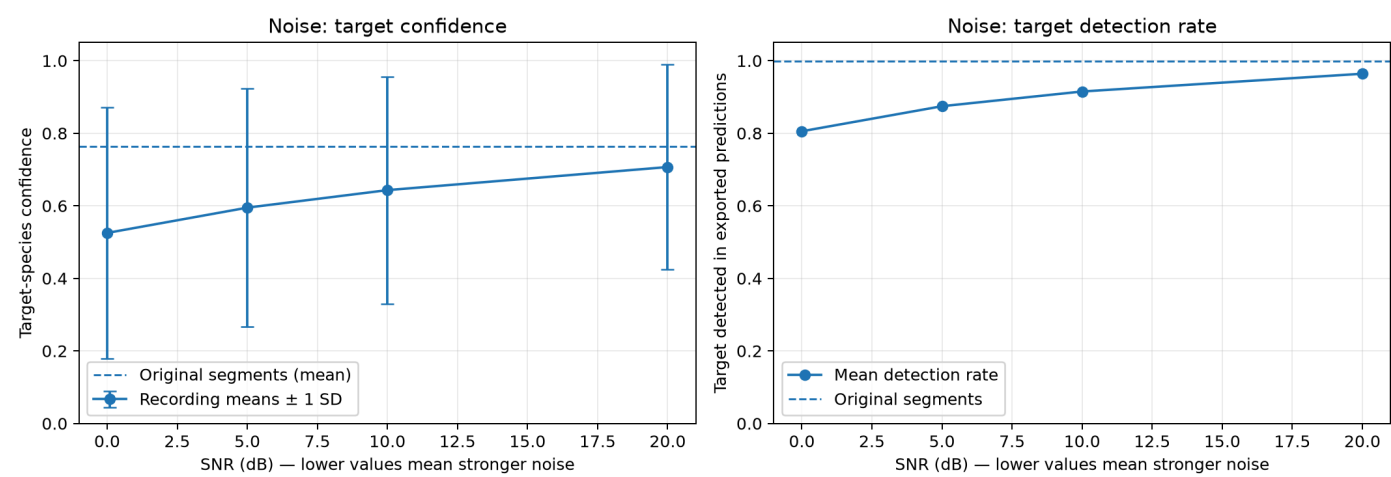
Data and coverage

2,005 recordings analysed · 95 failures · 3 completed batches · cursor 2,100

Corpus progress



Noise robustness



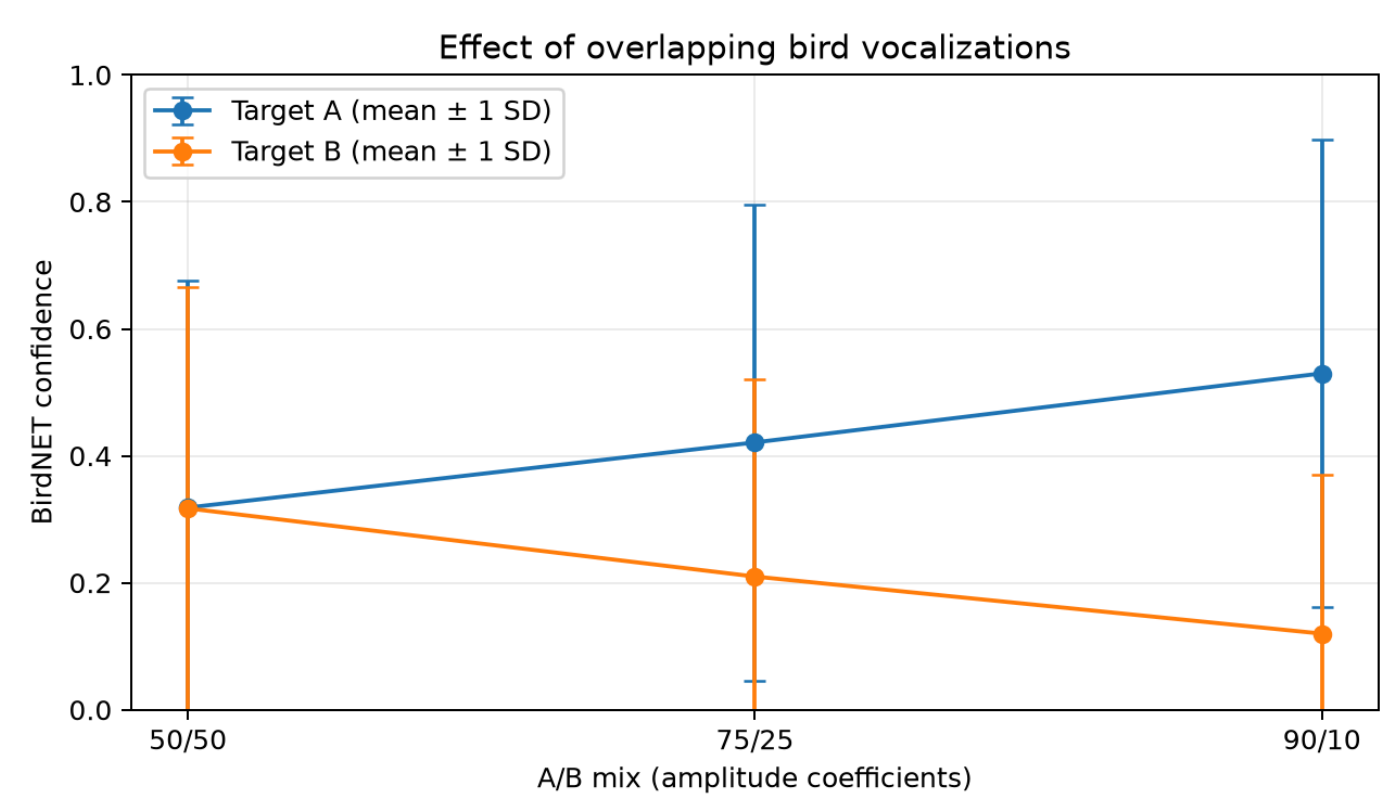
Observation

Segments from **1642 recordings** were tested. Mean target confidence without added noise is **0.764**.

- 20 dB**: mean confidence **0.707**, standard deviation across recordings **0.283**, target detected in **96.4%** of trials.
- 10 dB**: mean confidence **0.643**, standard deviation across recordings **0.313**, target detected in **91.5%** of trials.
- 5 dB**: mean confidence **0.595**, standard deviation across recordings **0.328**, target detected in **87.5%** of trials.
- 0 dB**: mean confidence **0.525**, standard deviation across recordings **0.346**, target detected in **80.5%** of trials.

Error bars show one standard deviation of recording means. A value of 0 means the target is absent from predictions exported at the minimum confidence threshold of 0.10. **363 recordings** without an initially detected target were excluded from this test.

Overlapping vocalizations

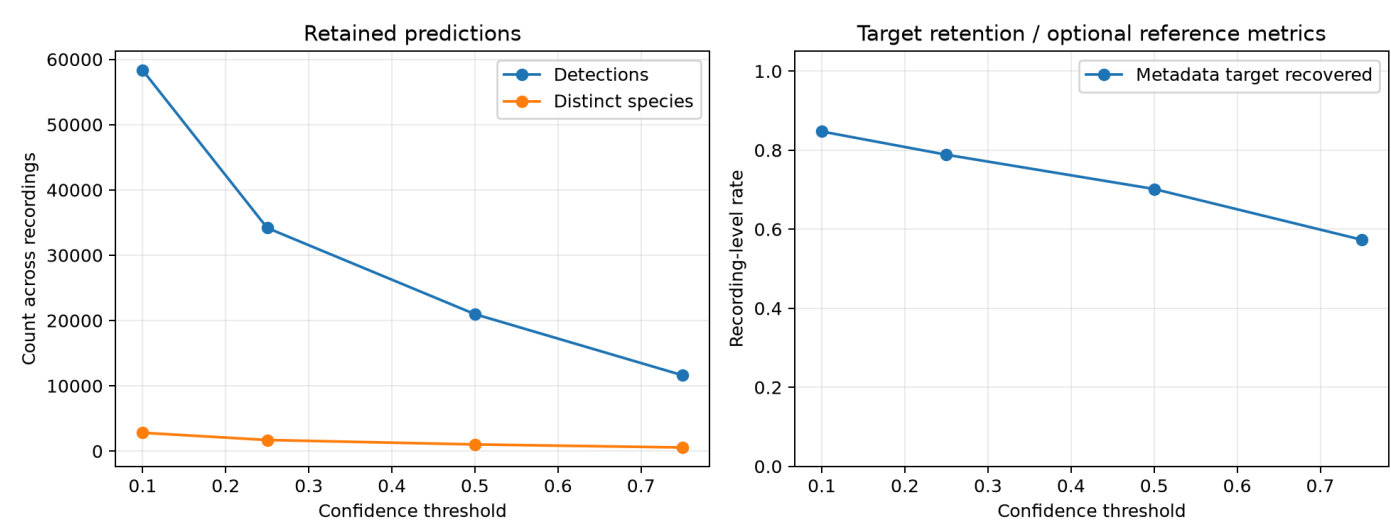


Observation

Means cover **798 pairs** of distinct recordings. Between mixes 50/50 and 90/10, mean target A confidence changes from **0.318** to **0.530**, and target B confidence from **0.317** to **0.120**.

A and B identify the roles within each pair. Error bars show one standard deviation across pairs. Coefficients apply to amplitude, without prior level equalization.

Confidence threshold



Observation

Between thresholds **0.10** and **0.75**, retained detections change from **58324** to **11612**, and predicted species from **2809** to **560**.

Xeno-canto's primary species is recovered in **84.7%** then **57.3%** of recordings whose target is identified and supported by BirdNET. This rate measures recovery of the known target; recall across all species requires complete annotations.

Precision, recall and F1 are not calculated without complete annotations.

Spectrogram

Example of a short analysed segment.

